VTPEH 6270 - Check Point 03: Data Exploration

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1 Introduction

1.1 Background

The EPA Toxics Release Inventory (TRI) tracks facility-reported toxic chemical releases to different environmental media (e.g., air and water) across the United States. TRI data are commonly used for environmental monitoring and to generate hypotheses about where releases may be concentrated and how releases enter the environment (U.S. Environmental Protection Agency 2026). Importantly, release amounts do not directly measure exposure or health risk, but pathway-specific release patterns can be a useful first step for identifying potential concerns that may warrant further investigation (U.S. Environmental Protection Agency 2026).

1.2 Research Question

Is the amount of toxic chemical releases associated with release pathway (air vs. water) across facilities in the United States in 2024?

In this report, the outcome variable **A** is the amount of toxic chemical releases (pounds), and the categorical variable **B** is release pathway (air vs. water).

1.3 Scientific Plausibility and Context

Air and water releases represent different environmental transport processes and potential exposure routes, so it is scientifically meaningful to compare release amounts by pathway. EPA documentation emphasizes that TRI reports releases by environmental medium and that these data can serve as a starting point for evaluating potential human health and environmental concerns when combined with other information (U.S. Environmental Protection Agency 2026).

Prior research using TRI-based measures has found that toxic air releases are associated with adverse infant health outcomes and may be more harmful than other release media such as water or land, supporting the idea that pathway differences matter for public health (Agarwal 2010). In addition, environmental justice research has shown that industrial hazards and related burdens are not evenly distributed across communities, motivating careful description of release patterns and pathways as an upstream indicator of potential inequities (Wilson et al. 2012).

2 Data Preparation

2.1 Import data

```r
# Packages
library(ggplot2)

# Set working directory
setwd("C:/Users/rui/Desktop/VTPEH 6270/2.19/Data")

# Load TRI data
tri <- read.csv("tri_2024_us.csv", check.names = FALSE)
```

2.2 Create a subset (<= 10 variables)

```r
# Create a subset
tri_sub <- tri[, c(
  "1. YEAR",
  "8. ST",
  "7. COUNTY",
  "23. INDUSTRY SECTOR",
  "37. CHEMICAL",
  "107. TOTAL RELEASES",
  "51. 5.1 - FUGITIVE AIR",
  "52. 5.2 - STACK AIR",
  "53. 5.3 - WATER",
  "65. ON-SITE RELEASE TOTAL"
)]

# Readable
names(tri_sub) <- c(
  "year", "state", "county", "industry_sector", "chemical",
  "total_releases", "fugitive_air", "stack_air", "water", "onsite_total"
)
```

2.3 Derived variables and pathway definition

For this research question, I define **air releases** as the sum of fugitive air emissions and stack air emissions. I then construct a long-format dataset with a categorical pathway variable (Air vs Water) to support grouped summaries and comparative visualizations.

```r
# Air releases
tri_sub$air_releases <- tri_sub$fugitive_air + tri_sub$stack_air

# Categorical pathway variable
tri_pathway <- data.frame(
  releases = c(tri_sub$air_releases, tri_sub$water),
  pathway = factor(
    c(rep("Air", nrow(tri_sub)), rep("Water", nrow(tri_sub))),
    levels = c("Air", "Water")
  )
)

# Log(1+x) transformation for right-skewed releases (keeps zeros)
tri_pathway$log_releases <- log1p(tri_pathway$releases)

summary(tri_pathway$releases)
summary(tri_pathway$log_releases)
```

Release amounts are highly right-skewed with many zero values (median = 0), so I used a log(1+x) transformation to support clearer visualization and comparison across pathways.

3 Grouped Summary Statistics

```r
# Mean, median, IQR by pathway (on log scale)
pathway_summary_raw <- aggregate(
  log_releases ~ pathway,
  data = tri_pathway,
  FUN = function(x) {
    c(
      mean = mean(x, na.rm = TRUE),
      median = median(x, na.rm = TRUE),
      IQR = IQR(x, na.rm = TRUE)
    )
  }
)

# Expand matrix column into separate columns
pathway_summary <- data.frame(
  pathway = pathway_summary_raw$pathway,
  mean_log = pathway_summary_raw$log_releases[, "mean"],
  median_log = pathway_summary_raw$log_releases[, "median"],
  IQR_log = pathway_summary_raw$log_releases[, "IQR"]
)

pathway_summary
```

Interpretation: On the log(1 + releases) scale, air releases are substantially higher and more variable than water releases. The median log release for air is 1.84 (IQR = 5.97), while the median and IQR for water are both 0, indicating that at least 75% of facility-records report zero water releases.

Overall, these grouped summaries suggest a strong association between release amount and release pathway, with air releases dominating in both magnitude and variability.

4 Comparative Visualizations

4.1 Histogram of releases by pathway (log scale)

```
p1 <- ggplot(tri_pathway, aes(x = log_releases)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ pathway, ncol = 1, scales = "free_y") +
  labs(
    title = "Distribution of Toxic Chemical Releases by Pathway",
    x = "log(1 + releases in pounds)",
    y = "Number of facility records"
  ) +
  theme_classic()

p1
```

```
ggsave(
  filename = "Figure1_Toxic_Releases_by_Pathway.pdf",
  plot = p1,
  device = "pdf",
  width = 180,
  height = 220,
  units = "mm"
)
```

Figure 1 compares the distributions of toxic chemical releases by pathway on a log-transformed scale. Air releases show a substantially wider distribution with a longer right tail, indicating greater variability and more frequent large release events compared with water releases.

In contrast, water releases are highly concentrated near zero, with relatively few facility-records reporting substantial discharges. These patterns suggest that the magnitude of toxic chemical releases is strongly associated with release pathway, with air pathways contributing disproportionately to higher release amounts across U.S. facilities in 2024.

5 AI Use Disclosure Statement

This document was generated with assistance from ChatGPT, which was used to help understand the assignment requirements and to debug and improve R code written by the author. All analyses, interpretations,and conclusions were made by the author.

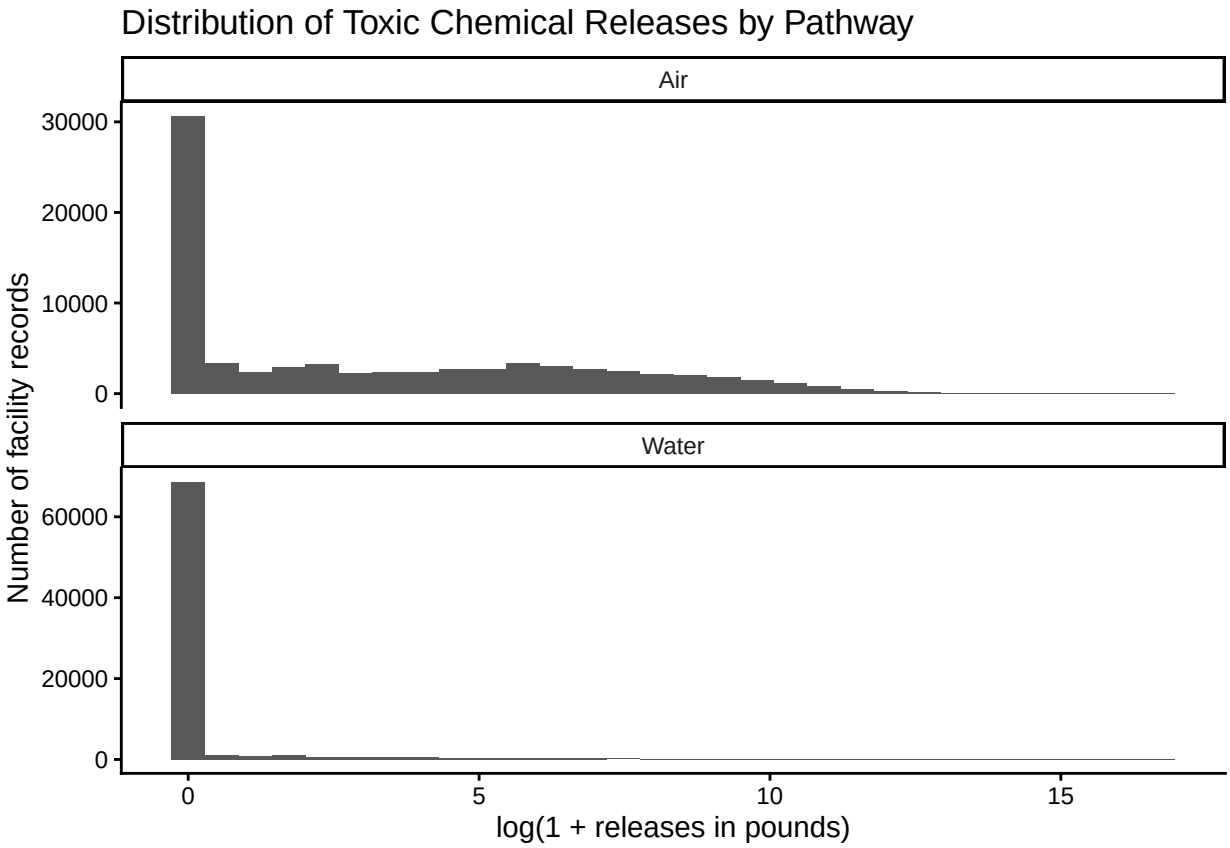

Figure 1: Distribution of toxic chemical releases by pathway (log1p scale).

References

Agarwal, N. 2010. "Toxic Exposure in America: Estimating Fetal and Infant Health Outcomes from 14 Years of TRI Reporting." *Journal of Health Economics* 29 (4): 557–74. https://doi.org/10.1016/j.jhealeco.2010.04.004.

U.S. Environmental Protection Agency. 2026. *TRI and Estimating Potential Risk*. https://www.epa.gov/toxics-release-inventory-tri-program/tri-and-estimating-potential-risk.

Wilson, Sacoby M., Halima Fraser-Rahim, Ebony Williams, Haorui Zhang, Lisa Rice, Erik Svendsen, and Willie Abara. 2012. "Assessment of the Distribution of Toxic Release Inventory Facilities in Metropolitan Charleston: An Environmental Justice Case Study." *American Journal of Public Health* 102 (10): 1974–80. https://doi.org/10.2105/AJPH.2012.300700.